Physics and Mathematical Formulation of the Active Multidomain Preisach Hysteresis Model

Krishna Chemudupati

April 1, 2026

Abstract

This note records the exact physics and mathematical structure used by the active multidomain Preisach hysteresis model in the `src/` tree. The purpose is to state precisely what is being solved in the current repository, rather than to survey the full range of possible ferroelectric models. The active workflow combines a multidomain Preisach constitutive law for AlScN, one-dimensional stack electrostatics with finite metal screening and a work-function-derived built-in term, and a bisection-based quasistatic self-consistency solver. The equations, parameters, and numerical protocol are written here in the same form used by the active implementation.

Contents

1 Problem statement

The active repository solves a quasistatic polarization–voltage hysteresis problem for stacks of the form

top metal | interlayer | AlScN | dead layer | bottom metal.

In the main active figures the dead layer is set to zero thickness, but the stack definition retains the dead-layer terms symbolically because they remain part of the implemented electrostatic model. The question answered by the active workflow is therefore narrow and concrete: for a fixed multilayer stack, a fixed domain population, and a prescribed voltage sweep, what self-consistent multidomain ferroelectric polarization is obtained at each bias point?

2 Variables and units

The macroscopic stack variables are the interlayer thickness t_i, the ferroelectric thickness t_f, the dead-layer thickness t_d, the dielectric constants k_i, k_f, and k_d, the top and bottom screening lengths λ_t and λ_b, the corresponding screening dielectric factors k_t and k_b, the top and bottom work functions W_t and W_b, the applied voltage V_a, the screening charge density σ_s, the dead-layer polarization P_d, and the net ferroelectric polarization P_{FE}.

The ferroelectric is represented as a population of N Preisach domains. Domain n is described by a structural parameter $\eta_n = (c/a)_n$, a local saturation polarization $P_{s,n}$, a local coercive field $E_{c,n}$, and a binary state variable $s_n \in \{-1, +1\}$. The net ferroelectric polarization is defined as

$$P_{\mathrm{FE}} = \frac{1}{N} \sum_{n=1}^{N} s_n P_{s,n}. \tag{1}$$

Internally, the code works in atomic units. This is handled in `src/atomicunits.py`, where voltage is converted from volts to atomic units, thickness is converted from nanometers to bohr, polarization is converted from μC/cm^2 to atomic units, and electric field is converted from MV/cm to atomic units. The electrostatic constant used internally is

$$\epsilon_0^{(\mathrm{au})} = \frac{1}{4\pi}. \tag{2}$$

All equations below are written in a form consistent with that implementation.

3 Multidomain Preisach constitutive law

The active model generates the ferroelectric domain population from a structural-disorder proxy, namely the local lattice ratio c/a. For each domain,

$$\eta_n \sim \mathcal{N}(\mu_{c/a}, \sigma_{c/a}). \tag{3}$$

In the active figure workflow the default values are

$$\mu_{c/a} = 1.54, \qquad \sigma_{c/a} = 0.04. \tag{4}$$

If no override values are supplied, the local saturation polarization and the local coercive field are generated through the default constitutive maps

$$P_{s,n} = 333.33\,\eta_n - 400 \qquad (\mu\mathrm{C/cm}^2\text{, before unit conversion}), \tag{5}$$

$$E_{c,n} = 3.16\,\eta_n - 1.1 \qquad (\mathrm{MV/cm}\text{, before unit conversion}). \tag{6}$$

The resulting values are then converted into atomic units. The class also supports an override path in which user-specified means and standard deviations for P_s and E_c are imposed by constructing linear maps from the sampled η_n values,

$$P_{s,n} = |a_P \eta_n + b_P| , \tag{7}$$
$$E_{c,n} = |a_E \eta_n + b_E| , \tag{8}$$

with

$$a_P = \frac{\sigma_{P_s}}{\sigma_{c/a}}, \qquad b_P = \mu_{P_s} - a_P \mu_{c/a}, \tag{9}$$
$$a_E = \frac{\sigma_{E_c}}{\sigma_{c/a}}, \qquad b_E = \mu_{E_c} - a_E \mu_{c/a}. \tag{10}$$

The absolute value in this override path is used only to prevent unphysical negative local magnitudes.

The switching law itself is a symmetric rectangular hysteron. If E denotes a candidate internal field seen by the ferroelectric, the domain update is

$$s_n(E) = \begin{cases} -1, & s_n^{\mathrm{base}} = +1 \text{ and } E \leq -E_{c,n}, \\ +1, & s_n^{\mathrm{base}} = -1 \text{ and } E \geq +E_{c,n}, \\ s_n^{\mathrm{base}}, & \text{otherwise.} \end{cases} \tag{11}$$

The current active model therefore assumes binary local states, abrupt threshold-driven switching, and no continuous local $P(E)$ law or explicit switching kinetics.

4 Stack electrostatics

The stack is instantiated by the `FerroelectricDiode` class and carries the geometry and boundary-condition data required by the electrostatic solver. The active implementation uses the thicknesses t_i, t_f, and t_d, the dielectric constants k_i, k_f, and k_d, the screening constants k_t and k_b, the screening lengths λ_t and λ_b, the work functions W_t and W_b, and the current polarization P_{FE}. In the current code the dead-layer parameters are

$$k_d = \frac{k_f}{2}, \qquad P_d = 0. \tag{12}$$

The work-function difference is not inserted as a single additive voltage through the whole stack. Instead, the code computes a ferroelectric-specific built-in voltage partition,

$$\Delta V_{\mathrm{bi,FE}} = (W_b - W_t) \frac{t_f}{t_f + \frac{k_f}{k_i} t_i + \frac{k_f}{k_d} t_d}. \tag{13}$$

This is the exact expression used in `src/potential.py`.

Given a polarization P_{FE} and an applied voltage V_a, the active code computes the screening charge density as

$$\sigma_s(P_{\mathrm{FE}}, V_a) = \frac{\frac{P_d t_d}{k_d} + \frac{P_{\mathrm{FE}} t_f}{k_f} + \epsilon_0 V_a}{\frac{\lambda_t}{k_t} + \frac{\lambda_b}{k_b} + \frac{t_d}{k_d} + \frac{t_f}{k_f} + \frac{t_i}{k_i}}. \tag{14}$$

The internal field inside the ferroelectric is then

$$E_{\rm FE}(P_{\rm FE}, V_a) = \frac{\sigma_s(P_{\rm FE}, V_a) - P_{\rm FE}}{k_f \epsilon_0} + \frac{\Delta V_{\rm bi,FE}}{t_f}. \tag{15}$$

Equation (14) says that the electrodes must supply enough free charge to compensate the external source term, the ferroelectric bound charge, and any dead-layer contribution while respecting the total series-capacitance-like denominator of the whole stack. Equation (15) then states that the ferroelectric feels a screening-controlled electrostatic field together with a built-in contribution set by the work-function asymmetry and the dielectric partition of the stack. This is the reason IL thickness, IL dielectric constant, metal work function, and metal screening length all influence the final hysteresis loop.

5 Self-consistency condition and bisection solver

At one applied voltage V_a, the solver first freezes the previous domain population $\{s_n^{\rm base}\}_{n=1}^{N}$. For a guessed field E, it builds a candidate switched state using (11) and then computes the corresponding candidate polarization,

$$P_{\rm FE}^{\rm cand}(E) = \frac{1}{N} \sum_{n=1}^{N} s_n(E) P_{s,n}. \tag{16}$$

The active bisection solver defines the residual

$$r(E; V_a) = E - E_{\rm FE}\Big(P_{\rm FE}^{\rm cand}(E), V_a\Big), \tag{17}$$

and seeks a field $E^\star$ such that

$$r(E^\star; V_a) = 0. \tag{18}$$

This equation is the appropriate quasistatic closure condition for the present model because it requires the field used to update the domains to be equal to the field that the stack electrostatics would actually produce for the polarization generated by those updated domains.

The active solver first builds an initial bracket by estimating a sufficiently large field window from the mean magnitude of the local polarization values, the extreme ferroelectric fields corresponding to $P = \pm P_{\rm max}$, and the maximum coercive-field magnitude in the domain population, then adds a margin of 1 MV/cm in atomic units. If the residual does not change sign across that interval, the bracket is expanded up to 24 times. This step is purely numerical; it is not additional physics. The bisection loop itself terminates when the polarization difference between the high-field and low-field candidates becomes sufficiently small,

$$\left| P_{\rm FE}^{\rm high} - P_{\rm FE}^{\rm low} \right| \le \Delta P_{\rm tol}, \tag{19}$$

and in the active figure workflow the tolerance is

$$\Delta P_{\rm tol} = 0.5\ \mu{\rm C/cm}^2. \tag{20}$$

If the bracket cannot be made sign-changing or if the iteration limit is reached, the solver chooses the candidate with the smallest absolute residual. That fallback is a numerical protection device rather than a new physical assumption.

6 Sweep protocol and active parameter set

The active figure driver constructs a single-cycle voltage trace

$$-V_{\max} \to +V_{\max} \to -V_{\max} \tag{21}$$

using a fixed voltage step, repeats that cycle twice,

$$\text{full sweep} = \text{cycle} \times 2, \tag{22}$$

and plots only the final cycle. This reduces the dependence of the displayed loop on the arbitrary initial domain state. Across the IL-thickness sweep the same seed and the same structural-distribution parameters are reused, so the same c/a realization and the same initial random domain-state pattern are preserved from one IL thickness to the next. This isolates the electrostatic effect of IL thickness rather than mixing geometry changes with random changes in the ferroelectric population.

The most important baseline material parameters used by the active figure path are summarized in Table 1.

Table 1: Baseline active material parameters used by the current figure workflow.

Material	Work function (eV)	Screening length (bohr)	k	Notes
Ti	4.33	0.1	1	baseline metal
Al	4.28	0.1	1	baseline metal
Pd	5.30	0.1	1	baseline metal
Test	4.08	1.0	1	lower WF, longer screening
Test2	4.33	1.0	1	same WF as Ti, longer screening
Al_2O_3	—	—	9.3	baseline AlO_x proxy
HfO_x	—	—	16.64	high-k interlayer
AlScN	—	—	16	baseline FE dielectric constant

The main thickness-specific voltage windows in the active figure set are 45 nm AlScN at ±20 V, 20 nm AlScN at ±12 V, and 10 nm AlScN at ±6.5 V.

7 Qualitative consequences of the equations

Equations (14) and (15) already predict the dominant trends in the figure set. Increasing the interlayer thickness raises the electrostatic denominator, reduces the charge-compensation efficiency of the electrodes, weakens the transmission of external voltage to the ferroelectric, and strengthens the depolarization penalty; as a result, the loop generally shrinks within a fixed voltage window. Increasing the interlayer dielectric constant has the opposite effect in the sense that a larger k_i reduces the electrostatic penalty associated with the interlayer. This is why the active model predicts larger loops for HfO_x than for Al_2O_3 at the same thickness.

The work-function difference enters through (13), so changing the top–bottom order changes the sign of the built-in term. Reversing a strongly asymmetric stack should therefore approximately reverse the horizontal preference of the loop. The screening lengths enter the denominator of (14) directly. Poorer screening reduces the screening charge available to compensate bound charge, strengthens the depolarization field, and can shrink both positive and negative branches even when the work function is unchanged.

8 Representative active results

The generated figure set is consistent with these expectations. Figures 1 and 2 show a representative comparison for the 20 nm AlScN case at ±12 V.

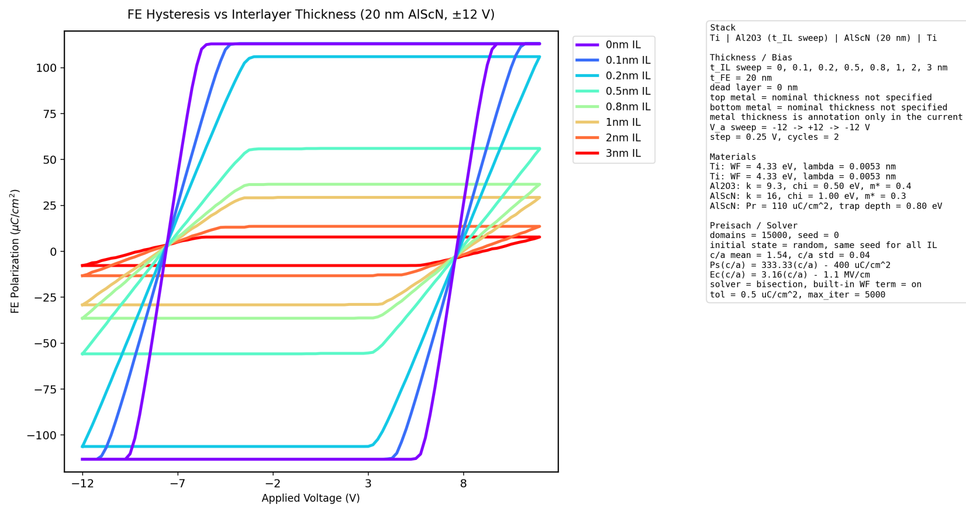

Figure 1: Representative Al_2O_3 interlayer sweep: 20 nm AlScN at ±12 V.

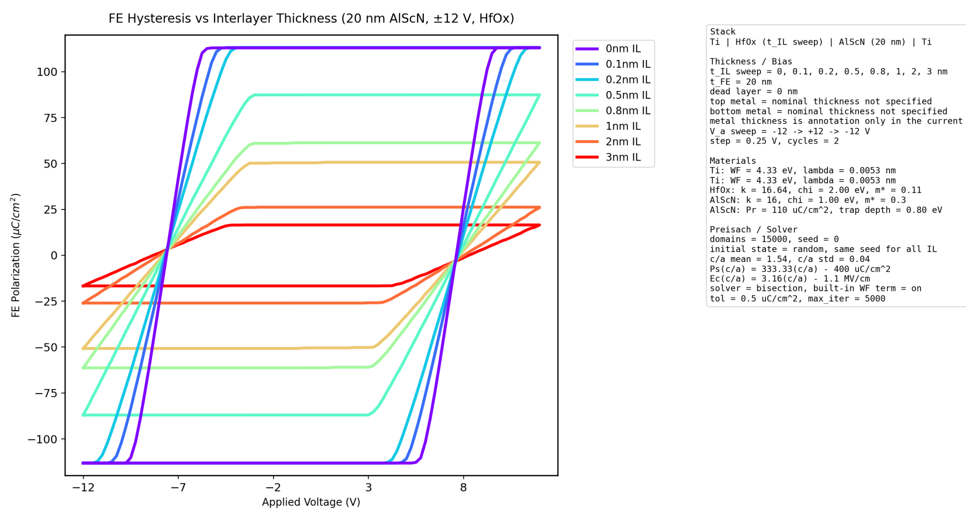

Figure 2: Representative HfO_x interlayer sweep: 20 nm AlScN at ±12 V.

For the 20 nm case at 3 nm IL, the Al_2O_3 stack gives $P_{\max} = 7.830\ \mu\mathrm{C/cm}^2$, whereas the HfO_x stack gives $P_{\max} = 16.533\ \mu\mathrm{C/cm}^2$. This is the expected electrostatic trend associated with the larger dielectric constant of HfO_x.

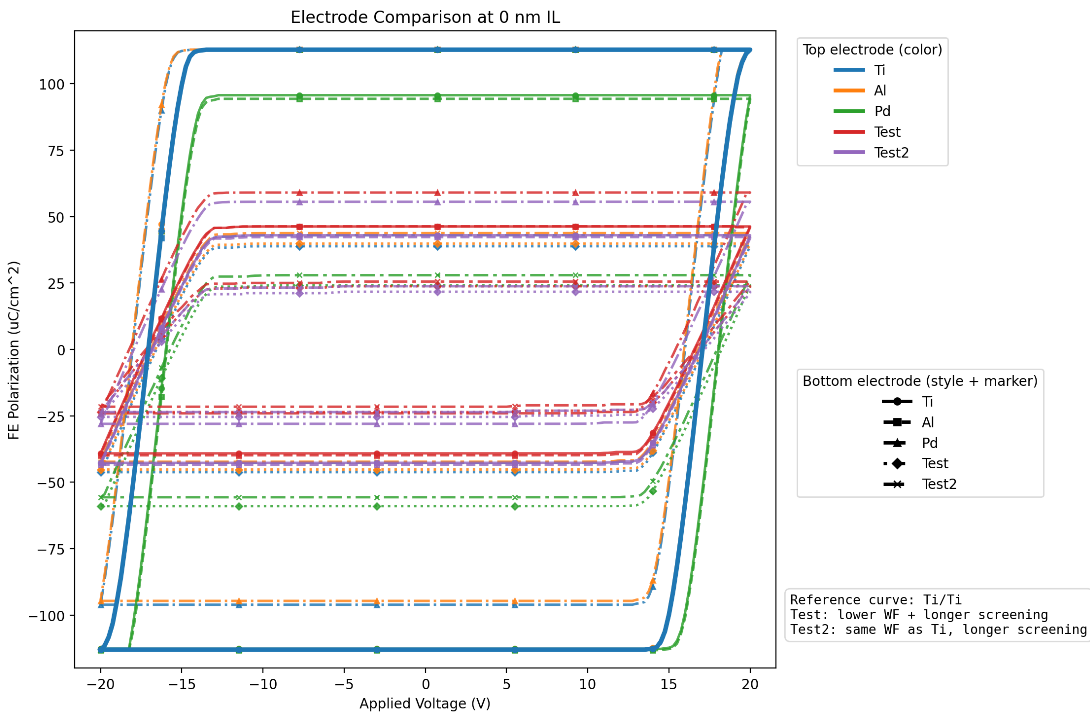

Figure 3: Electrode sensitivity at 0 nm IL.

The electrode comparison provides a second useful check. Ti/Test2 differs strongly from Ti/Ti even though Test2 preserves the Ti work function. This is a direct signature of screening-length sensitivity in the active electrostatic model, and it confirms that the metal boundary condition is controlled by screening as well as work-function asymmetry.

9 Scope of the active model

The present model is multidomain, history-dependent, electrostatically self-consistent, quasistatic, and physically interpretable. It is not a transport model, a time-domain switching-kinetics model, a continuous Landau or phase-field model, or a microscopic *ab initio* domain simulation. Those statements are not incidental; they define the domain of validity of the active workflow. The repository is solving a compact, interpretable hysteresis problem rather than a full diode transport problem.

10 Closing statement

The active repository solves the following problem: for a fixed multilayer stack and a fixed Preisach domain population, find the field-dependent domain configuration whose net polarization is self-consistent with the one-dimensional stack electrostatics at each point of a quasistatic voltage sweep. That statement captures the exact physical and mathematical content of the current multidomain Preisach hysteresis workflow.